

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103**

Summer Semester(Regular) Examination May – 2019

Branch:- Computer Science and Engineering

Subject :- Object Oriented Programming Using C++

Subject Code : - BTCOE404B

Semester :- IV

Marks:- 60

Date:- 22/5/2019

Duration :- 3 Hours.

Instructions to the Students:

1. Attempt **any five** questions of the following and each question carry 12 Marks.
2. Figures to the right indicates full marks.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.
5. Assume suitable data wherever necessary and mention it clearly.

	Marks
Q. 1 Solve the following.	12
A) a) Distinguish between the following terms:	6
i) Objects and Classes	
ii) Data abstraction and Data encapsulation	
iii) Inheritance and Polymorphism	
B) What is the role of constructor in C++? Explain various types of constructors with example.	6
Q.2A) What is operator overloading? Write the rules for operator overloading.	6
B) Write a menu-driven program for addition, subtraction, display result of two distances (Given in meter and centimeter) using member functions and friend function.	6
Q.3A) How is Polymorphism achieved at a) compile time b) run time? What is the difference between run time binding and compile time binding?	6
B) Explain virtual function with suitable example	6
Q.4A) Enlist header files in C++.Draw and explain stream class hierarchy.	6
B) Explain 'this' pointer with simple program example	6
Q.5A) What is an exception? How is an exceptions handle in C++? What are the advantages of using exceptions handling mechanism in a program?	6
B) Write a program in C++ by using swap () function from the functional template.	6
Q.6A) What are the components of STL? Explain them.	6
B) Write the following any two short notes	6
i) Iterators	
ii) Container adaptors	
iii) Vector	

END

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103

WINTER SEMESTER (Supplementary) EXAMINATION Dec.-2019

Branch: B.Tech. (Computer Engineering)

Sem: IV

Subject: Object Oriented Programming in C++[BTCOE404B]

Marks:60

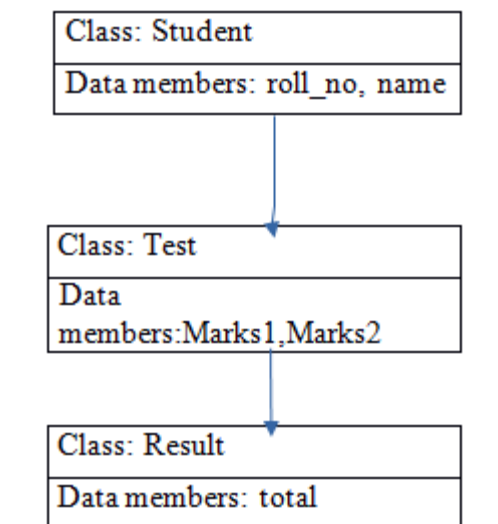
Date:- 02/12/2019

Time: 3 Hrs

Instructions to the Students:

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

	Marks
Q.1 Solve following questions.	
(A) Mention advantages of object oriented approach over functional/procedural programming.	06
(B) How do you define class in oop? Write a c++ program that will ask for a temperature in Fahrenheit and display it in Celsius.(formula= $((f-32)/1.8)$)	06
Q.2 Attempt the following questions.	
(A) Why is it necessary to overload an operator? Write a c++ program to overload '+' operator.	06
(B) Implement inheritance using following figure with member functions for reading and printing data.	06



Q.3 Solve any Two.

(A) Define virtual function. Compare compile time polymorphism and run time polymorphism with suitable real time application. 06

(B) What are the application of this pointer?explain with help of c++ program. 06

(C) Write a c++ program to read data from file and display 06
a)Number of lines in file
b)number of words in file
c)size of file

Q.4 Attempt the following questions.

(A) A template is considered as kind of macro. Then what is difference between them. 06

(B) How is exception handled in c++? Elaborate your answer with array index out of bound exception. 06

Q.5 Attempt the following questions.

(A) Define the term virtual base class and its implementation in c++.How it is used in function overriding. 06

(B) Write a c++ program to declare class mobile having data members as price and model number. Accept and display this data for five objects. 06

Q.6 Attempt the following questions.

(A) Differentiate standard template library and c++ standard library. 06

(B) Elaborate concept of vector in c++ with suitable c++ program. 06

*****End of Paper*****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch :Computer Engineering Semester :III Subject Code & Name: BTCOC305(A) Object Oriented Programming in C++ Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	State the need of object oriented programming and any two characteristics of C++.	Remember	6
B)	What is constructor? Explain Parameterized constructor.	Understanding	6
C)	Explain objects as function arguments in C++ with suitable example.	Apply	6
Q.2	Solve Any Two of the following.		12
A)	Write a C++ program for Binary operator overloading for addition of two complex numbers.	Apply	6
B)	Explain data conversion for primitive data type to user-defined type with C++ program.	Understanding	6
C)	Explain different types of Inheritance in C++.	Remember	6
Q. 3	Solve Any Two of the following.		12
A)	What are Virtual functions? How it is differ from simple functions?	Analysis	6
B)	Illustrate this pointer with the help of C++ program.	Apply	6
C)	What is friend function? State its uses.	Remember	6
Q.4	Solve Any Two of the following.		12
A)	Explain Classes for I/O and File stream operations.	Remember	6
B)	What are manipulators in C++?	Understanding	6
C)	Write a C++ program to write contents in a file.	Apply	6
Q. 5	Solve Any Two of the following.		12
A)	What is exception? State its need of Exception handling.	Remember	6
B)	Explain function template overloading with C++ program.	Apply	6
C)	Explain sequence type container in C++.	Understanding	6
*** End ***			